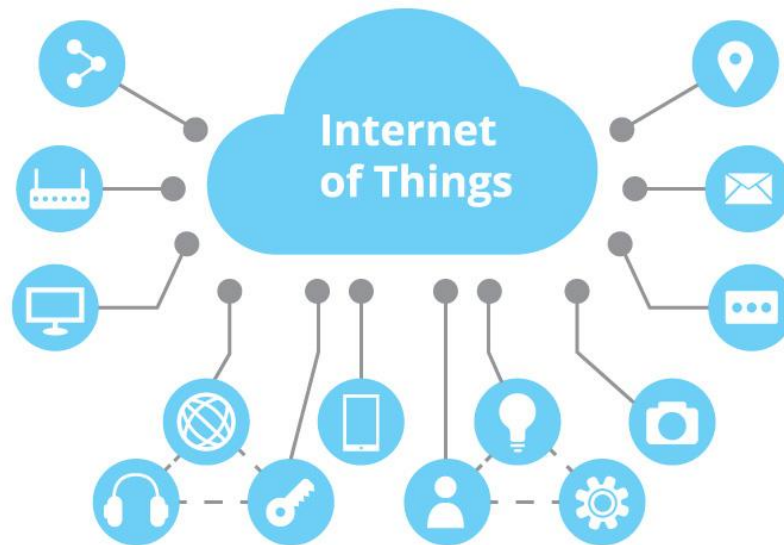


Internet of Things (IoT)



Chapter 4: Building an IoT System

Nguyen Tan No

04/2025

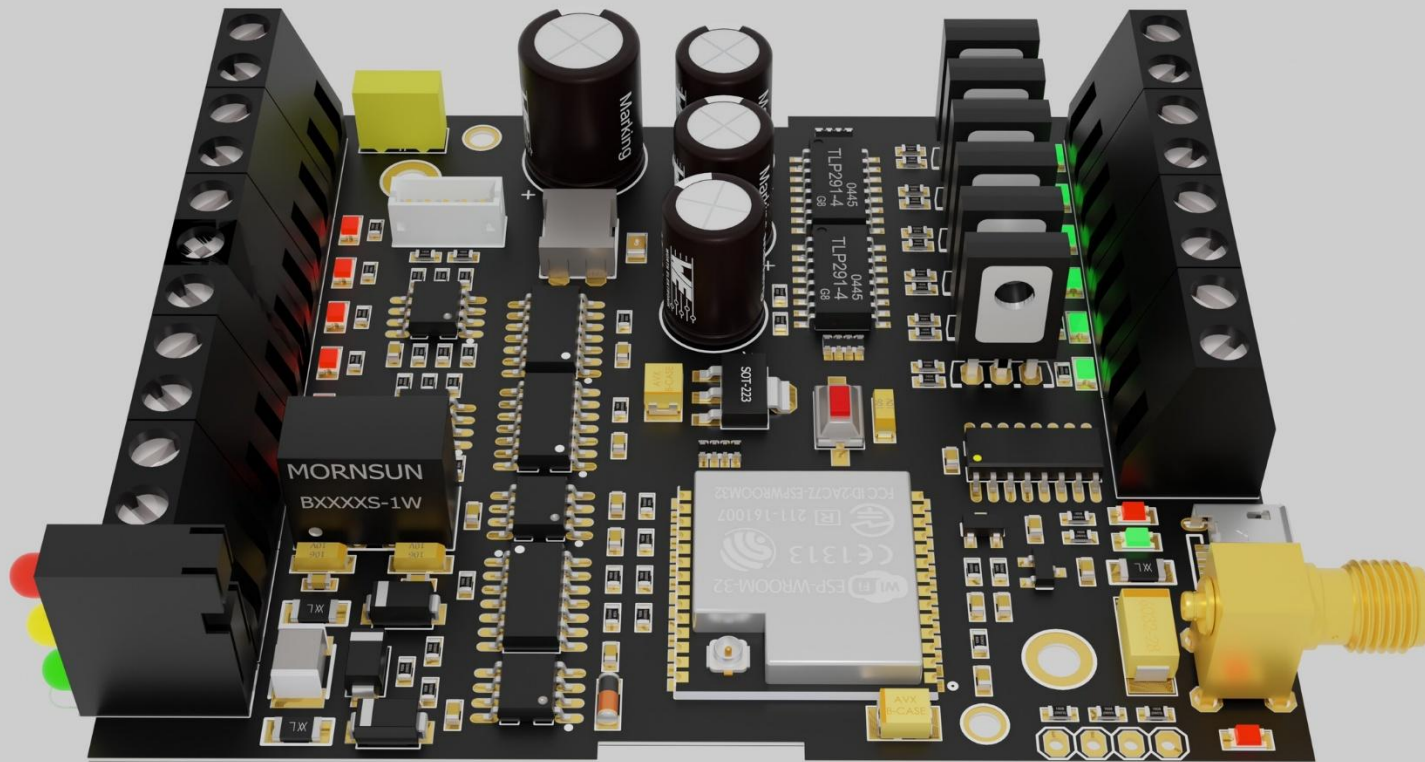
Contents

4.1. Hardware-Sensor Connection

4.2. Hardware-Actuator Connection

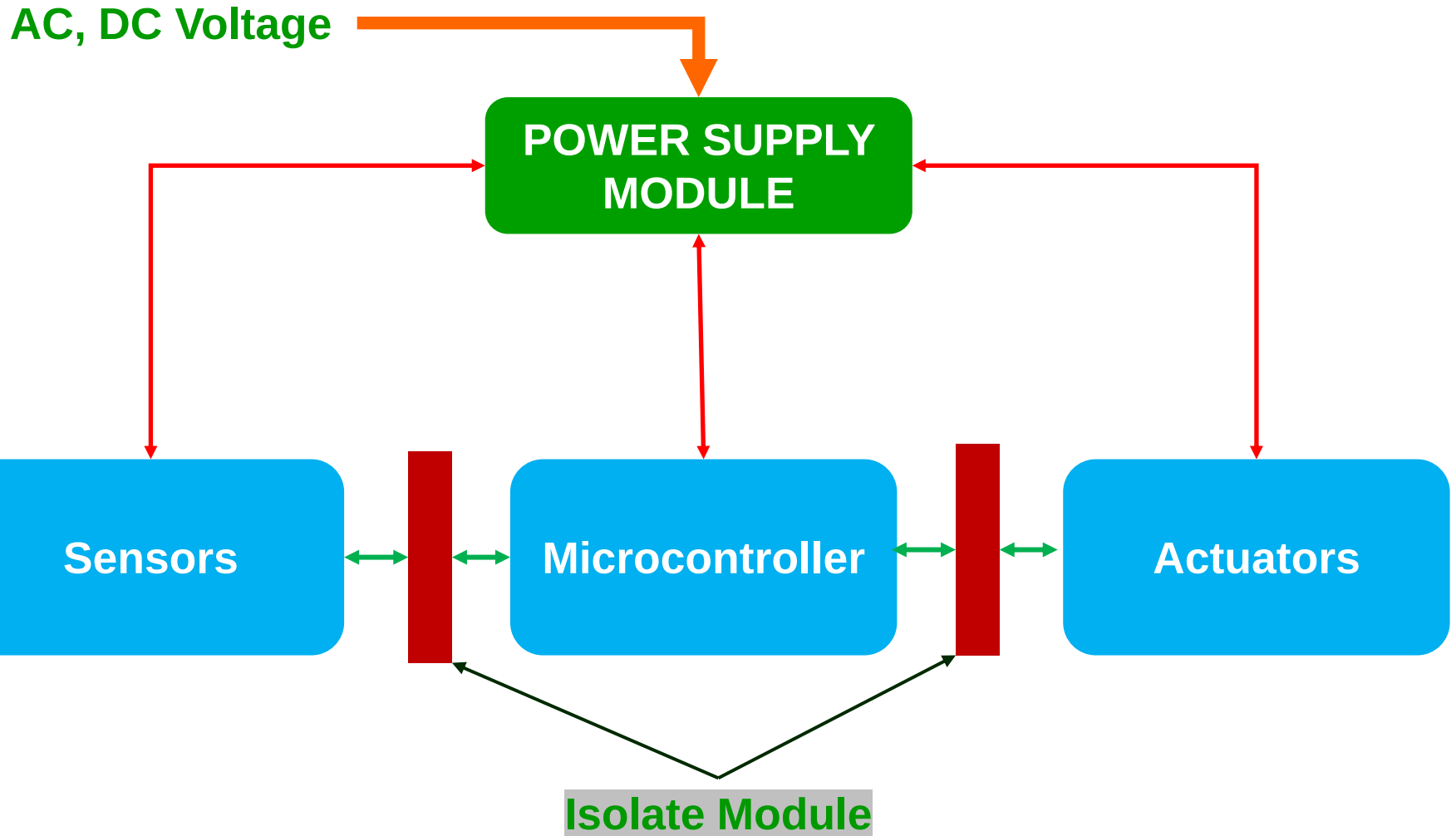
4.3. Programming cloud connectivity

4.1. Hardware-Sensor Connection

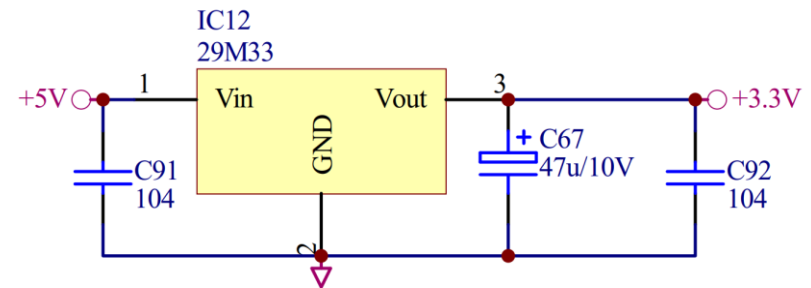
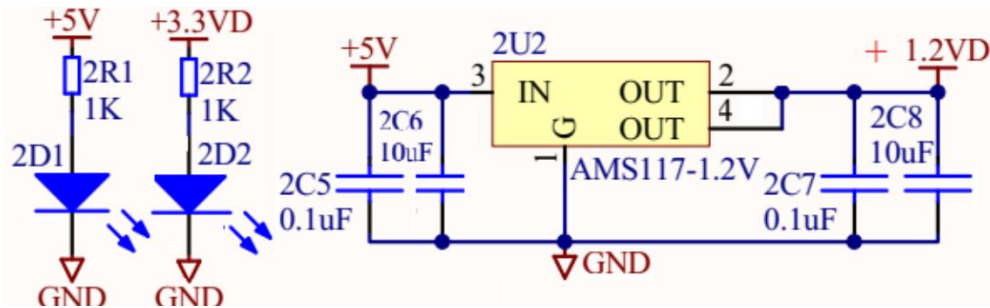
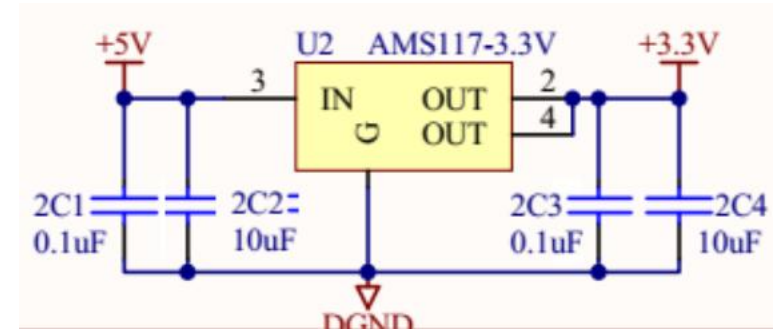
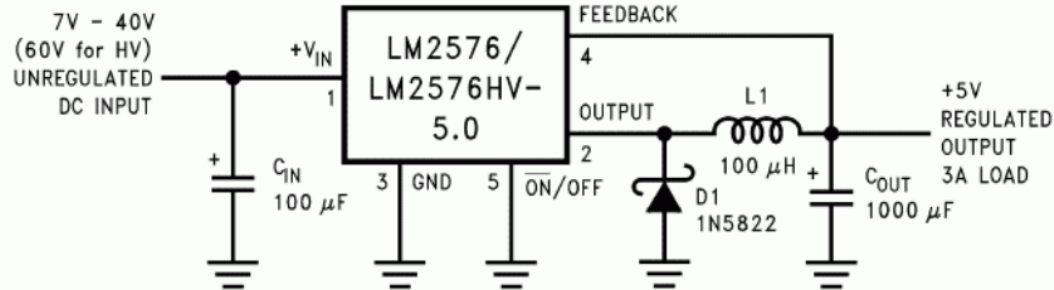


IoT Module

4.1. Hardware-Sensor Connection

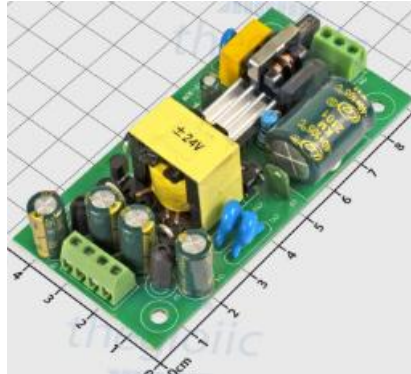


4.1. Hardware-Sensor Connection



Power Supply Shematic

4.1. Hardware-Sensor Connection



AC-DC



7805

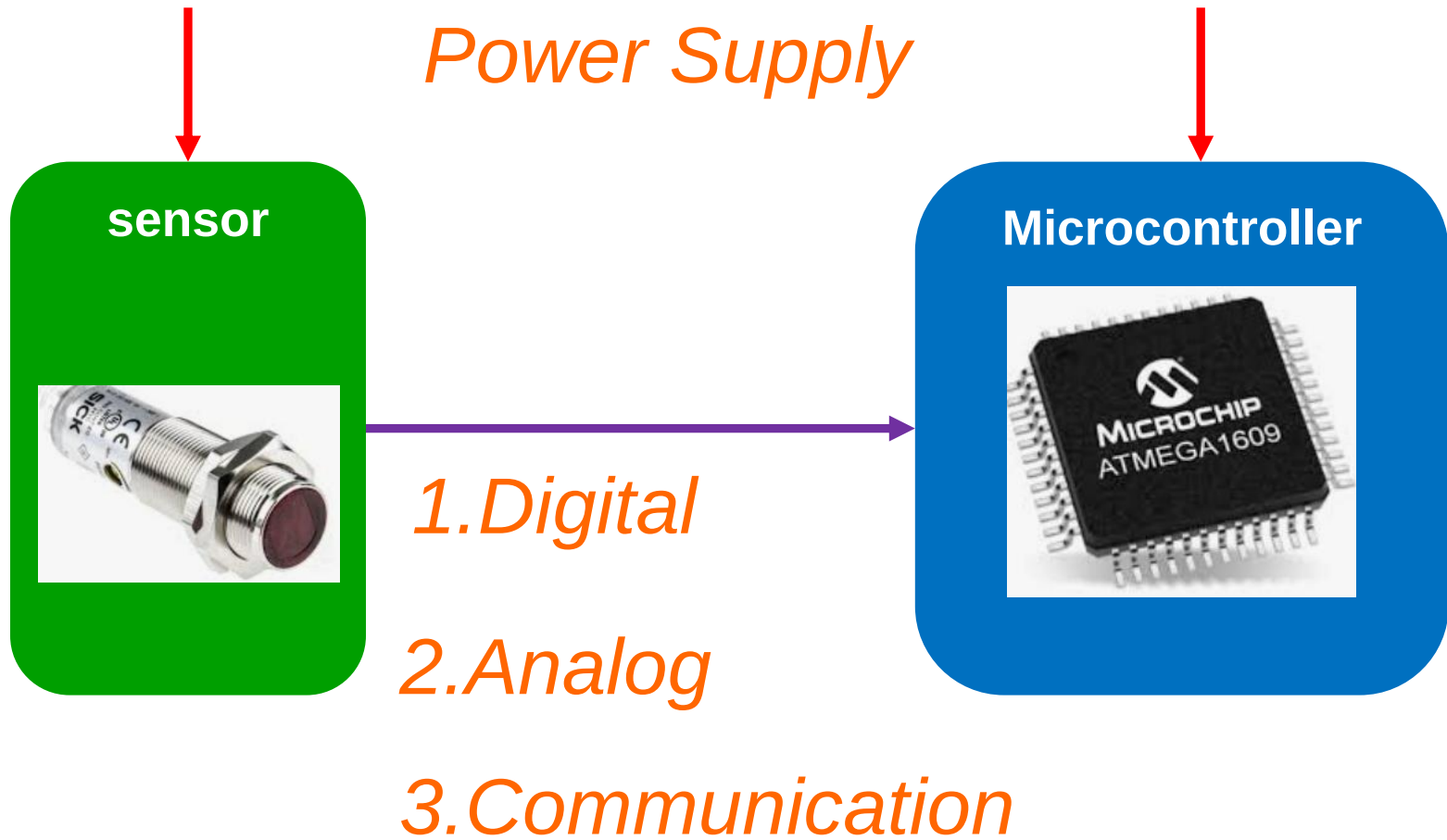


LM2576



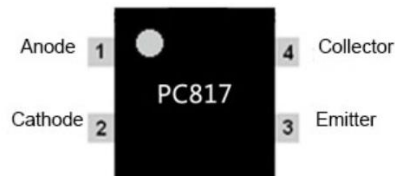
AMS1117

4.1. Hardware-Sensor Connection

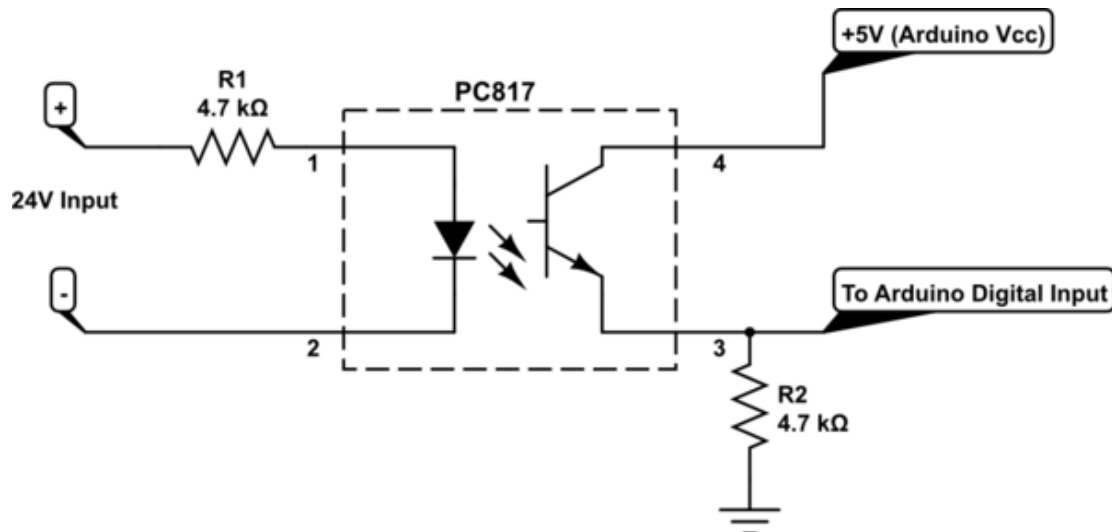


4.1. Hardware-Sensor Connection

1. Digital

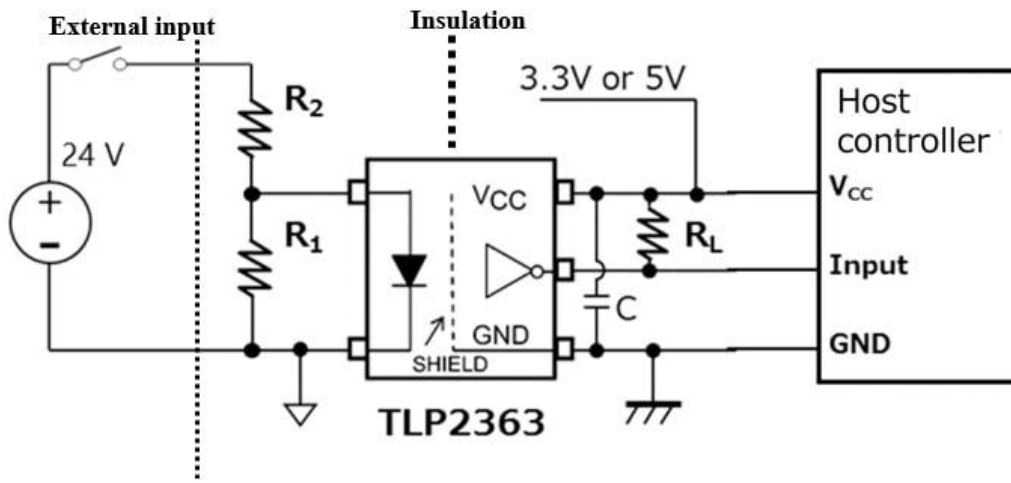


Parameter		Symbol	Conditions
Input	Forward voltage	V_F	$I_F = 20\text{mA}$
	Peak forward voltage	V_{FM}	$I_{FM} = 0.5\text{A}$
	Reverse current	I_R	$V_R = 4\text{V}$
	Terminal capacitance	C_t	$V = 0, f = 1\text{kHz}$
Output	Collector dark current	I_{CEO}	$V_{CE} = 20\text{V}$
Transfer characteristics	*Current transfer ratio	CTR	$I_F = 5\text{mA}, V_{CE} = 5\text{V}$
	Collector-emitter saturation voltage	$V_{CE(sat)}$	$I_F = 20\text{mA}, I_C = 1\text{mA}$
	Isolation resistance	R_{iso}	DC500V, 40 to 60%RH
	Floating capacitance	C_f	$V = 0, f = 1\text{MHz}$
	Cut-off frequency	f_c	$V_{CE} = 5\text{V}, I_C = 2\text{mA}, R_L = 100\Omega, -3\text{dB}$
Response time	Rise time	t_r	$V_{CE} = 2\text{V}, I_C = 2\text{mA}, R_L = 100\Omega$
	Fall time	t_f	

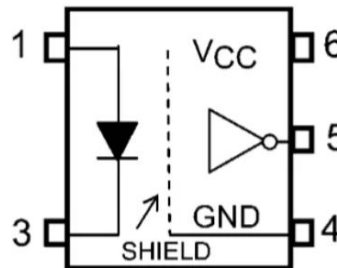
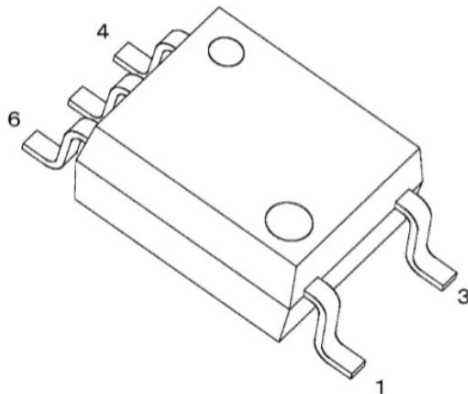


4.1. Hardware-Sensor Connection

1. Digital

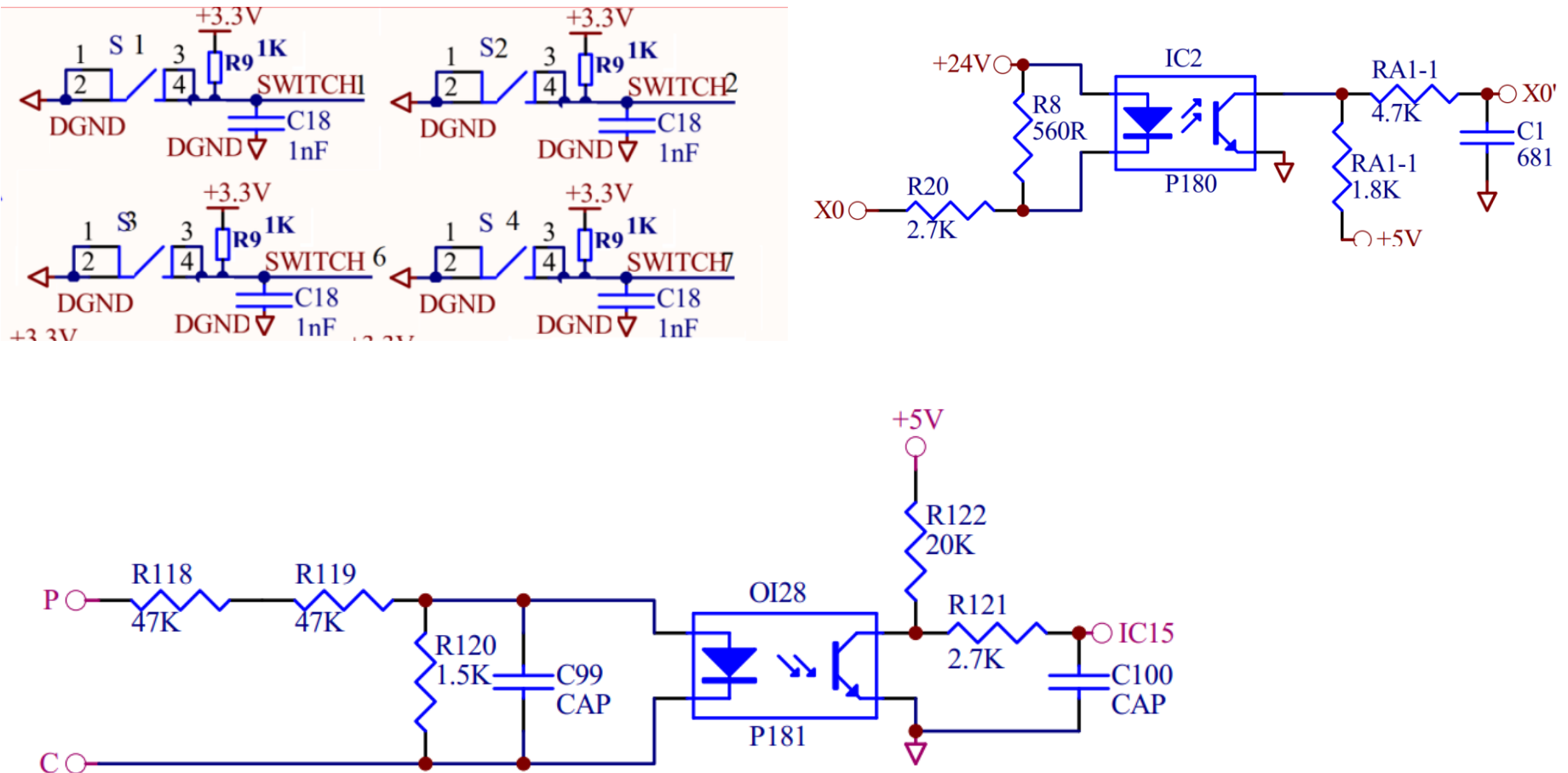


- Inverter logic type (open collector output)
- SO6 packaging
- -40°C to $+105^{\circ}\text{C}$ operating temperature range
- 2.7V to 5.5V supply voltage
- 10Mbps (typ.) data transfer rate
- 4mA (max) supply current
- 0.3mA to 2.4mA ($V_{CC} = 3.3\text{V}$, $R_L = 1\text{k}\Omega$) threshold
- $\pm 20\text{kV}/\mu\text{s}$ (min) common-mode transient immunity
- 3750V_{RMS} (min) isolation voltage



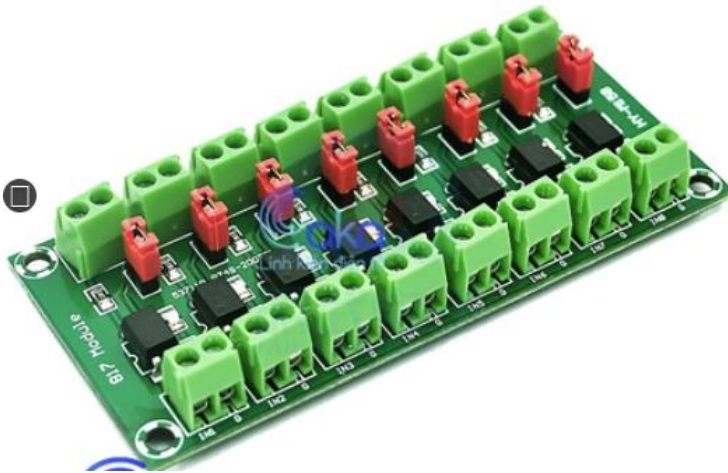
- 1: Anode
- 3: Cathode
- 4: GND
- 5: V_O (Output)
- 6: V_{CC}

4.1. Hardware-Sensor Connection



Input module shematics

4.1. Hardware-Sensor Connection

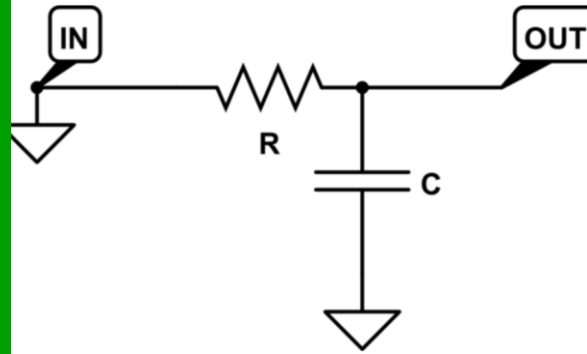


Opto Module

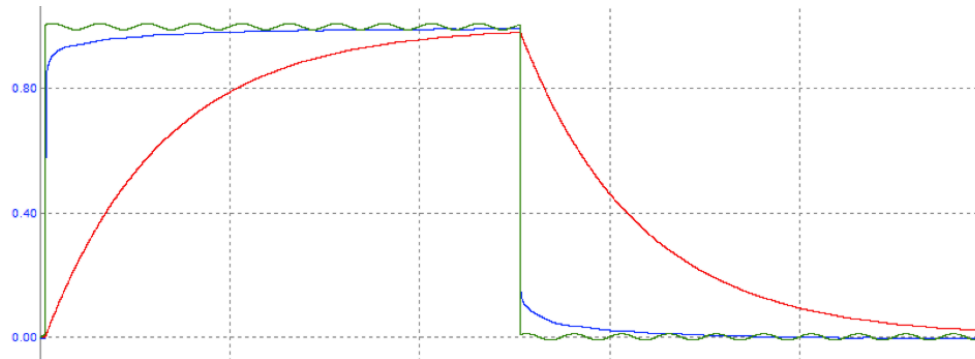
4.1. Hardware-Sensor Connection

2. Analog

sensor

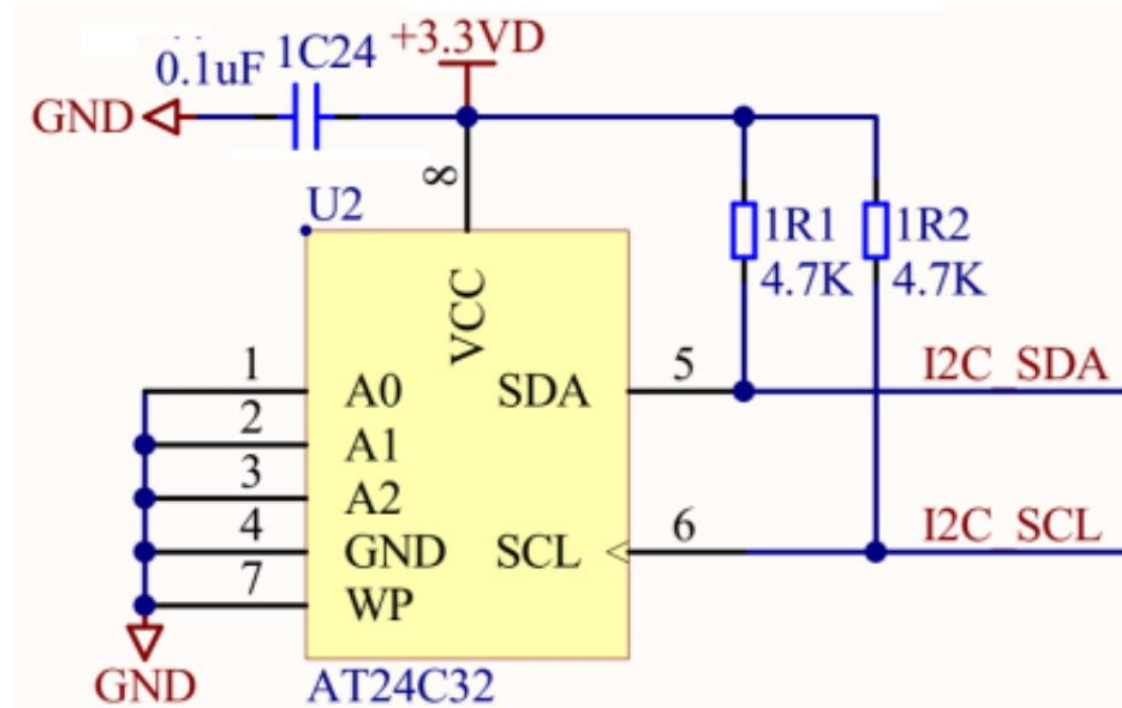


Microcontroller



4.1. Hardware-Sensor Connection

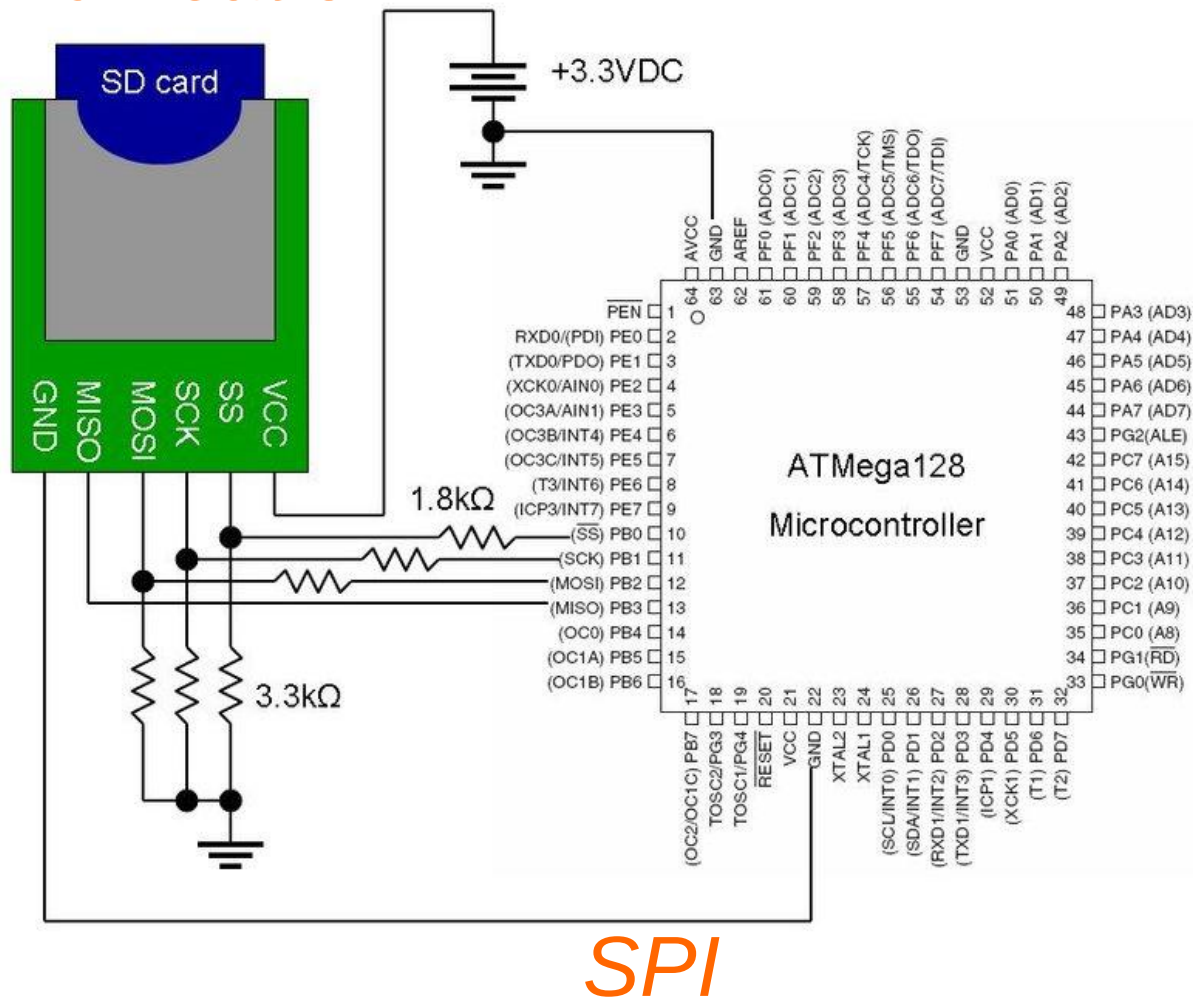
3.Communication



IIC

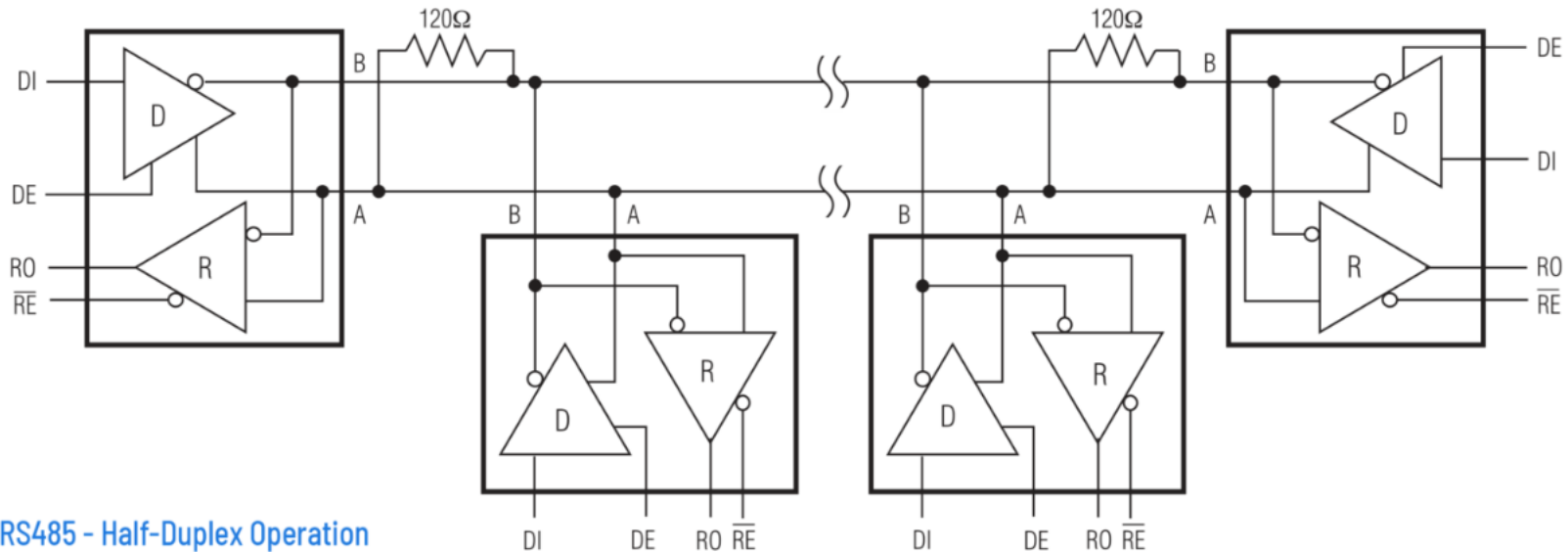
4.1. Hardware-Sensor Connection

3.Communication

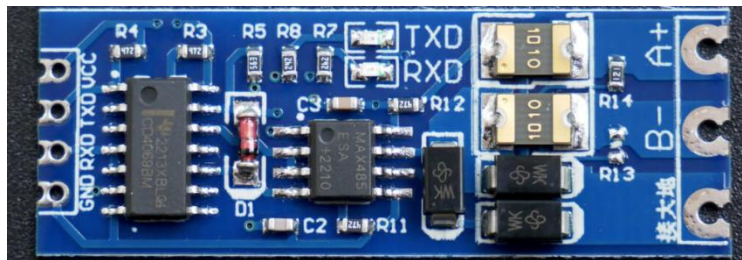


4.2. Hardware-Sensor Connection

3.Communication

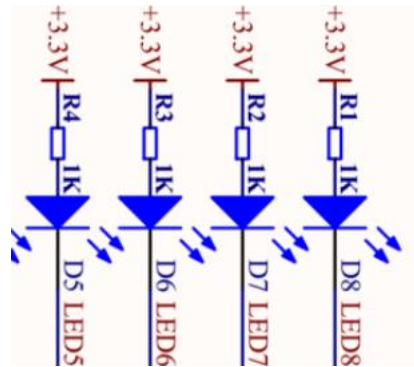


RS485 - Half-Duplex Operation

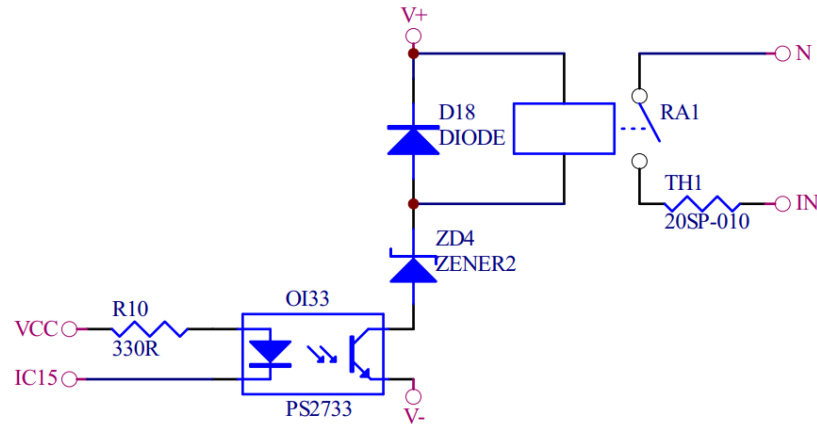


RS485

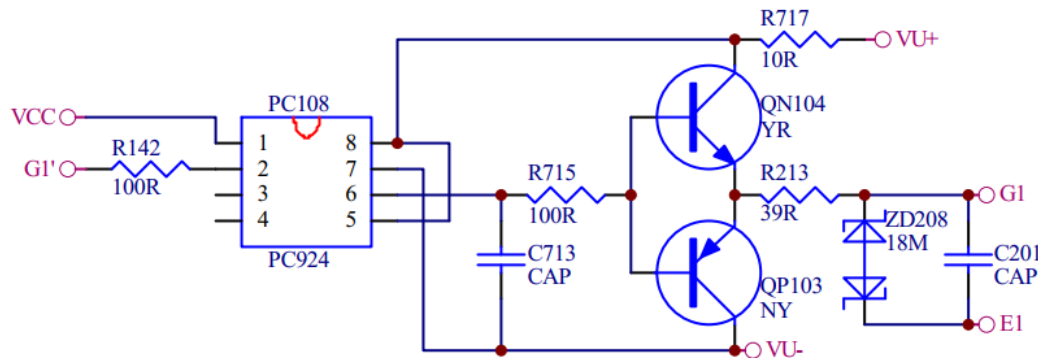
4.2. Hardware-Actuator Connection



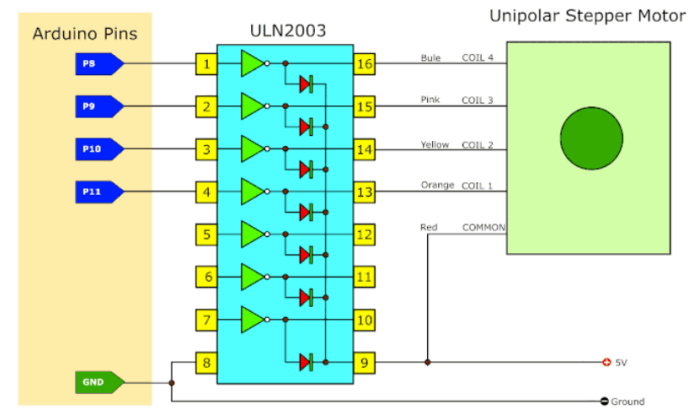
Led



Relay

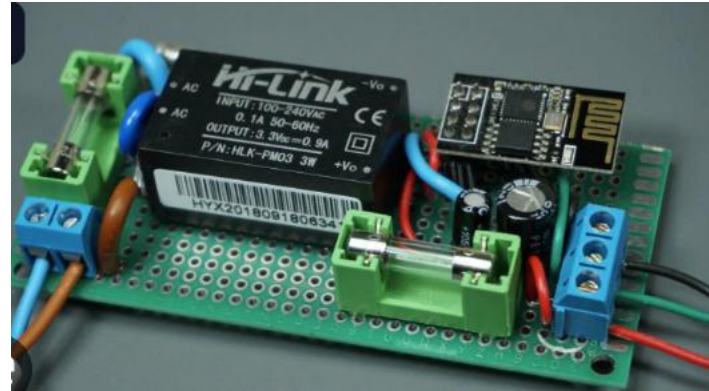
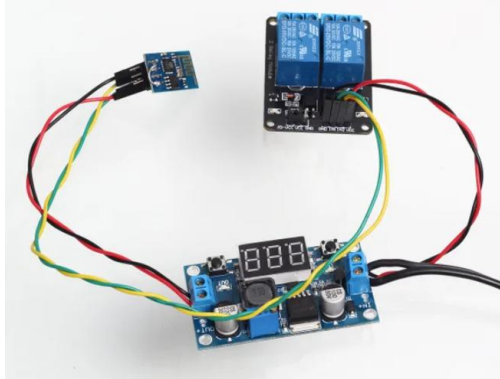


Transistor



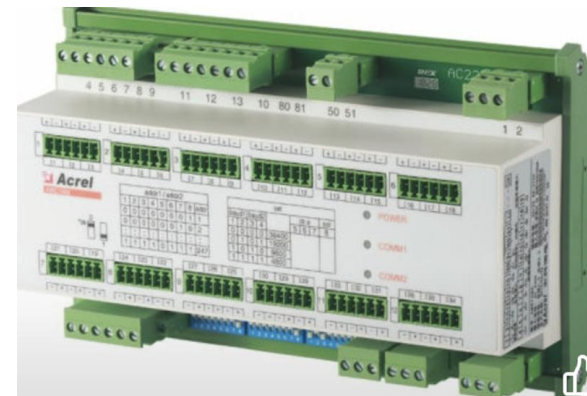
IC

4.2. Hardware-Actuator Connection



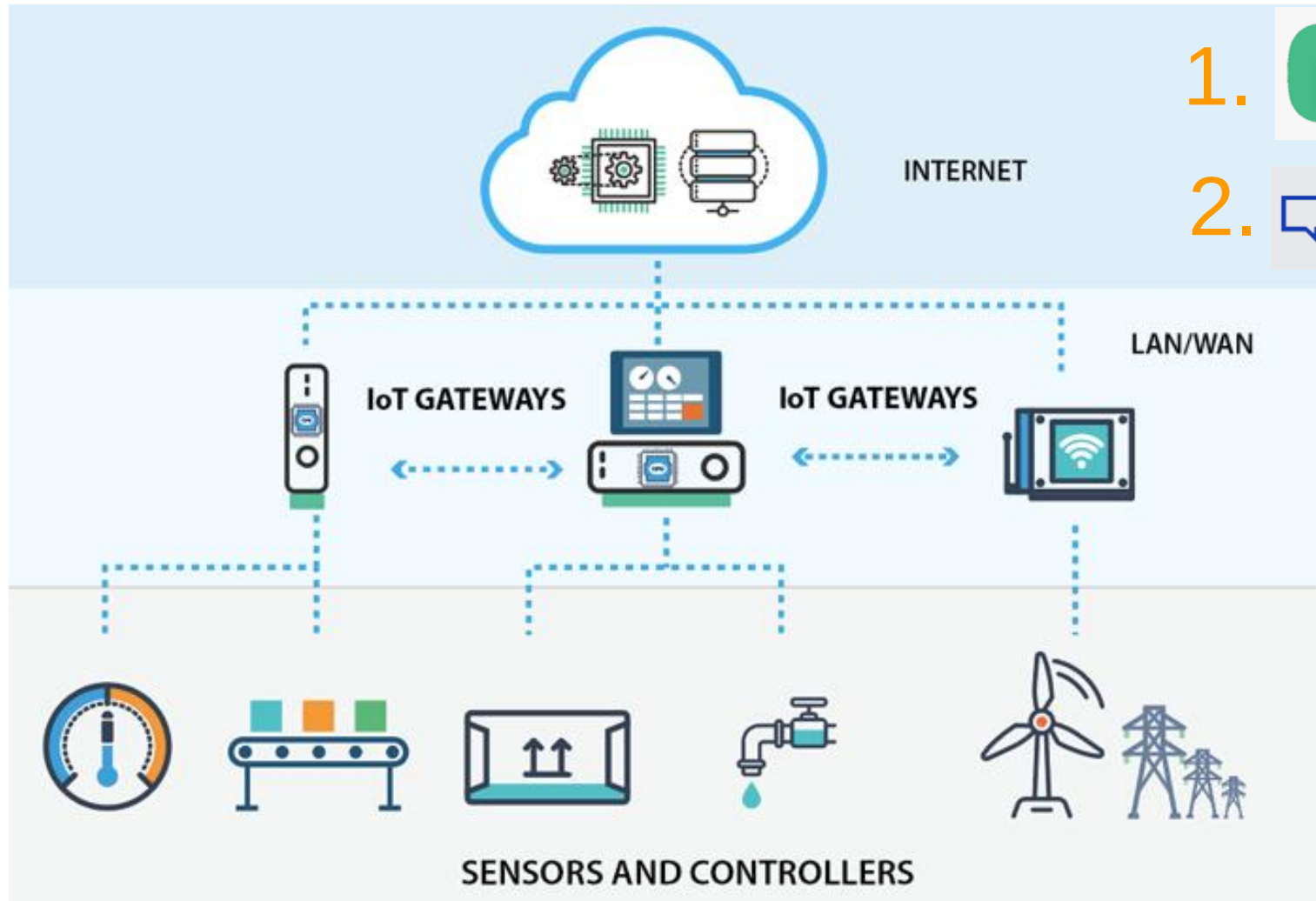
How to connect modules

4.2. Hardware-Actuator Connection



Product packaging

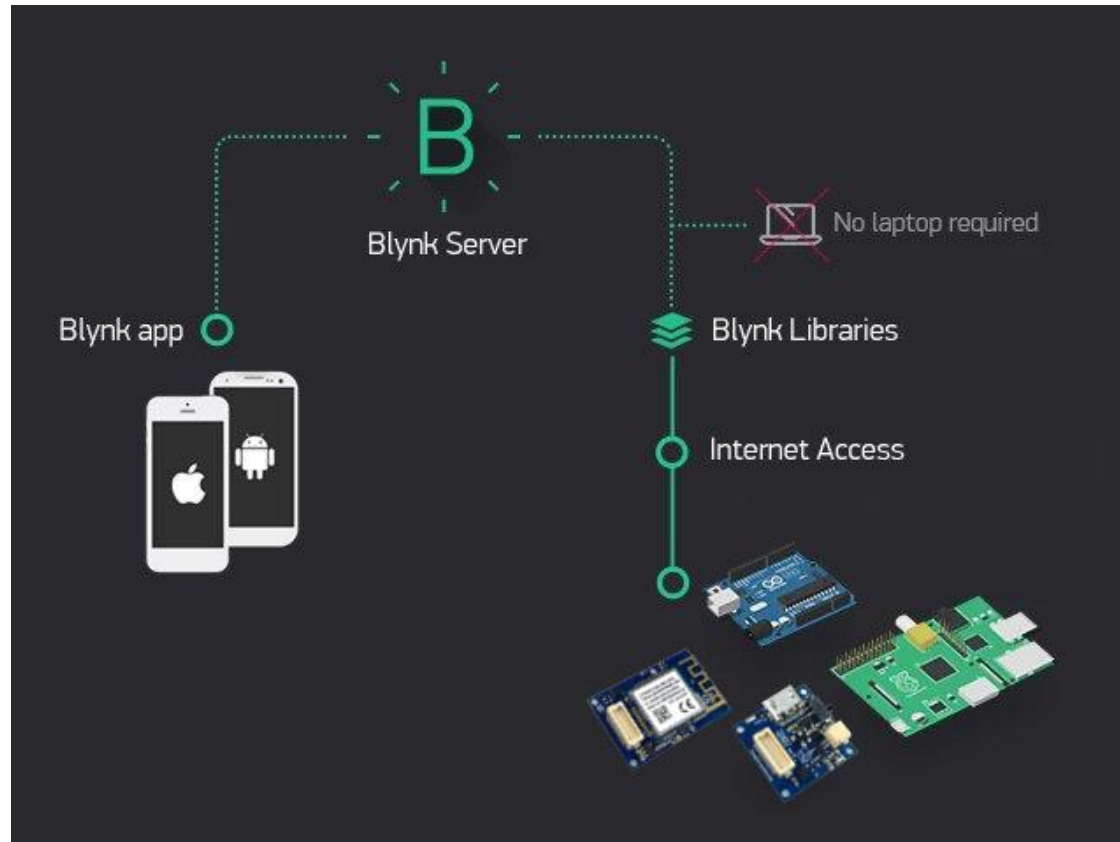
4.3. Programming cloud connectivity



1.  Blynk

2.  ThingSpeak

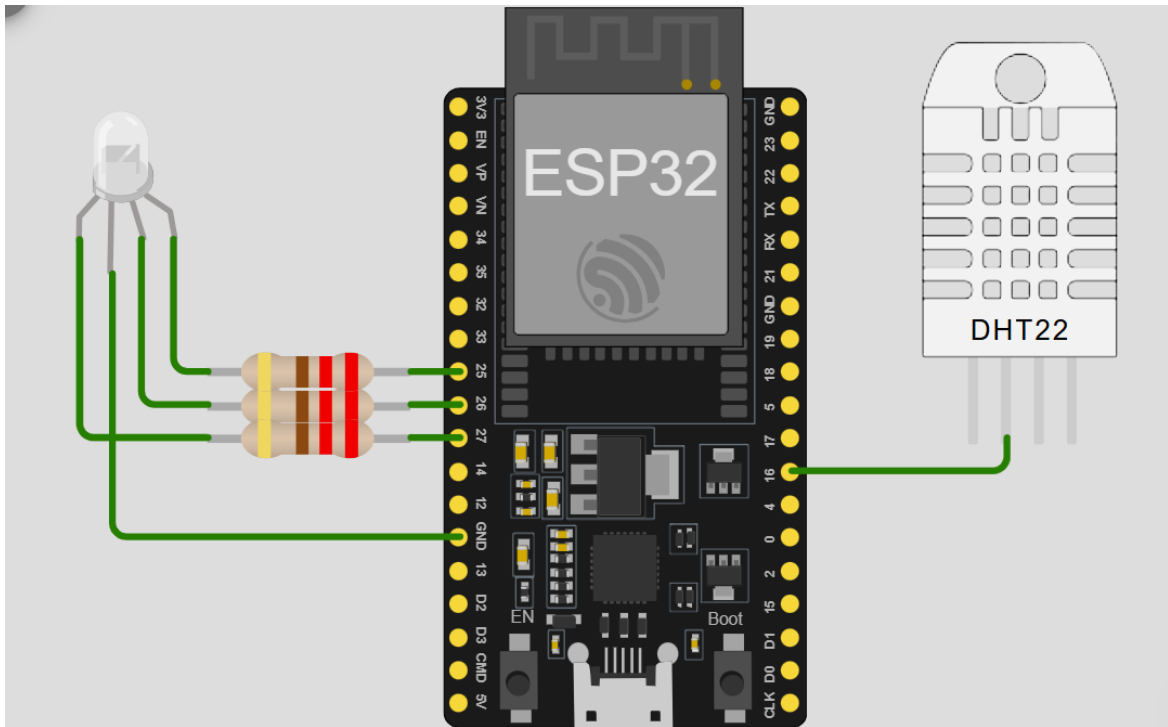
4.3. Programming cloud connectivity



"Blynk is an IoT (Internet of Things) platform that helps users easily create remote control applications for smart devices. What makes Blynk special is its flexibility and ease of use."

4.3. Programming cloud connectivity

Project 1:



4.3. Programming cloud connectivity

The screenshot displays the Blynk Developer Zone interface. The left sidebar contains navigation links: Developer Zone (1), Devices, Automations, Users, Organizations, and Locations. The main area is divided into 'My Templates' and 'Datastreams' sections. The 'My Templates' section includes a search bar and a list of templates: Blueprints, Blynk.Air (OTA) (2), Static Tokens, Rule Engine, OAuth 2.0, Webhooks, and Integrations. A 'Test' button (3) is located below the list. The 'Datastreams' section features a search bar and a table of datastreams (4). The table has columns for Id, Name, Pin, Color, and Data Type. It lists two datastreams: 'Double V0' (5) and 'Integer V1'. The right side of the interface shows a 'Device Name' (Online) with a dashboard (7) containing two widgets: 'Double V0 (V0)' (6) and 'Integer V1 (V1)'. The bottom right section includes a 'Blynk by Volodymyr Shymanskyi' banner with an 'Installation' section showing version 1.3.2 and an 'Add to Project' button.

Developer Zone 1

My Templates 2

Search Templates

Test 1 Device 3

Datastreams 4

Search datastream

Id	Name	Pin	Color	Data Type
1	Double V0	V0	Blue	Double
2	Integer V1	V1	Green	Integer

5

Device Name Online

Device Owner Company Name

Dashboard 7

Double V0 (V0) 6

Integer V1 (V1)

Blynk by Volodymyr Shymanskyi

Build a smartphone app for your project in minutes. Blynk allow Works with over 400 boards like ESP8266, ESP32, Arduino, Rasp

Installation

1.3.2 released 2 years ago

Add to Project | More info

4.3. Programming cloud connectivity

Home

1 Devices

PRO

Device name	Status	Auth Token
esp32	Offline	hXH7 - - -

What's next?

- ☐ Configure template
- ☐ Set up the Web Dashboard

Done:

- ☒ Set Up Datastreams
- ☒ Add first Device

Template settings

ESP32, WiFi



Firmware configuration



Template ID and Template Name should be declared at the very top of the firmware code.

```
#define BLYNK_TEMPLATE_ID  
"TMPL6gE_zhsqw"  
#define BLYNK_TEMPLATE_NAME "Test"
```

```
1  #define BLYNK_TEMPLATE_ID  
2  #define BLYNK_TEMPLATE_NAME  
3  #define BLYNK_AUTH_TOKEN  
  
5  #include <WiFi.h>  
6  #include <WiFiClient.h>  
7  #include <BlynkSimpleEsp32.h>  
8  char ssid[] = "Wokwi-GUEST";  
9  char pass[] = "";
```

```
"TMPL6gE_zhsqw"  
"Test"  
'hXH7SulBBqnDCltIdjaKgiedn_H_-h7w'
```


4.3. Programming cloud connectivity

```
23 void setup()  
24 {  
25   Serial.begin(115200);  
26   Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);  
27   timer.setInterval(1000L, myTimerEvent);  
28 }  
29 void loop()  
30 {  
31   Blynk.run();  
32   timer.run();  
33 }
```

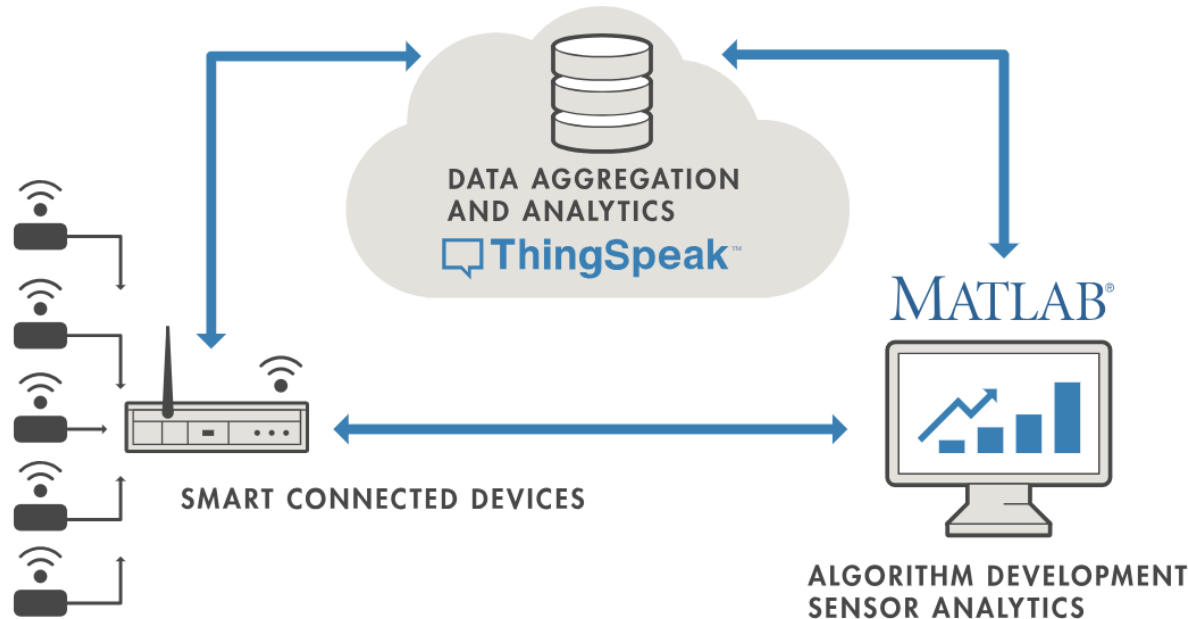
Read

```
BLYNK_WRITE(V1)  
{  
  int pinValue = param.asInt();  
  Serial.println(pinValue);  
}
```

Write

```
void myTimerEvent()  
{  
  counter+=3.4;  
  Serial.println(counter);  
  Blynk.virtualWrite(V0, counter);  
}
```

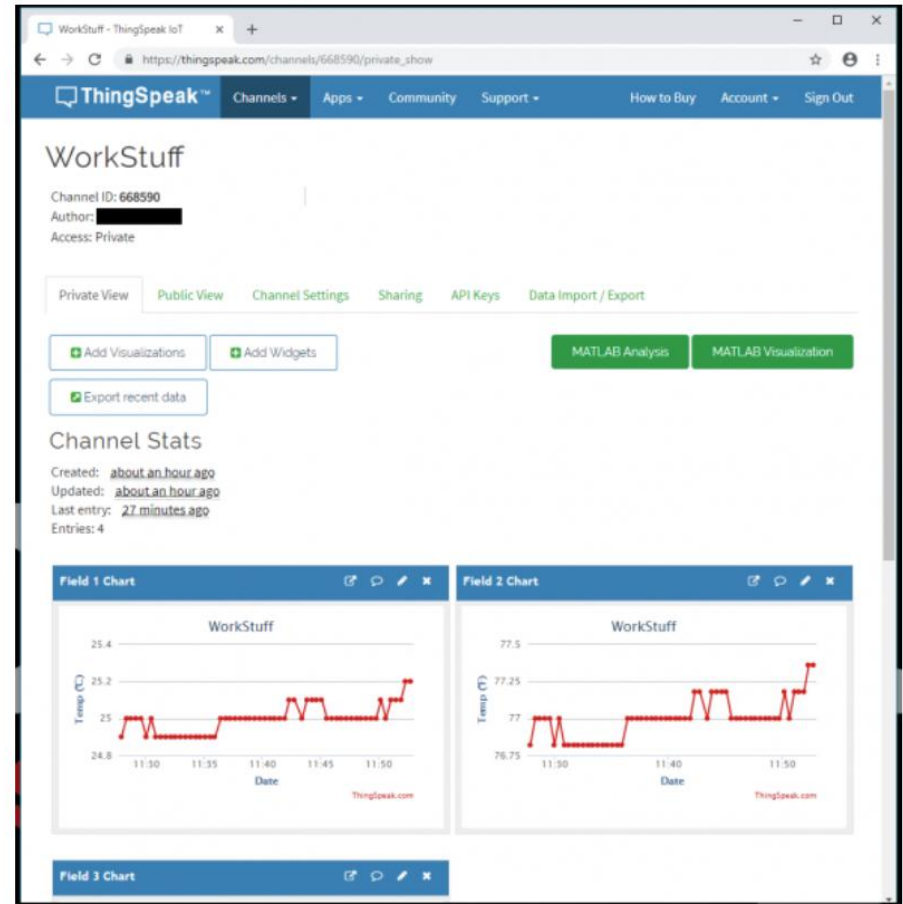
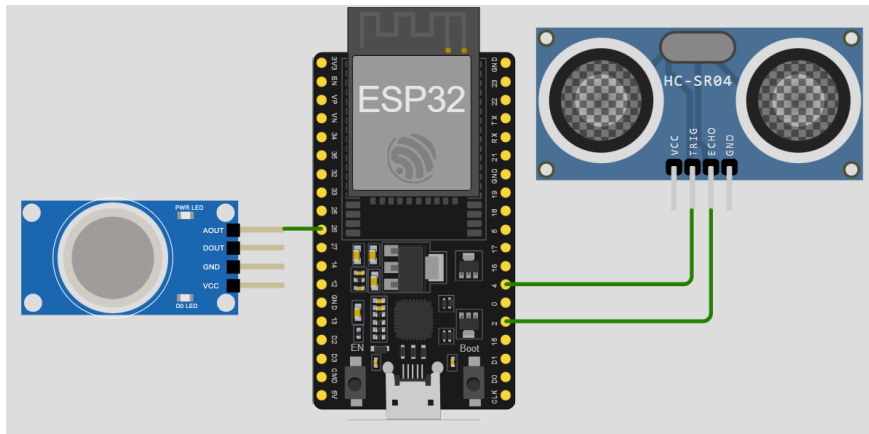

4.3. Programming cloud connectivity



ThingSpeak™ is an IoT analytics platform service that allows you to aggregate, visualize and analyze live data streams in the cloud. ThingSpeak provides instant visualizations of data posted by your devices to ThingSpeak. With the ability to execute MATLAB® code in ThingSpeak you can perform online analysis and processing of the data as it comes in. ThingSpeak is often used for prototyping and proof of concept IoT systems that require analytics.

4.3. Programming cloud connectivity

Project 2:



4.3. Programming cloud connectivity

thingspeak.mathworks.com/channels

ThingSpeak™ Channels Apps Devices

New Channel 1

Search by tag

Name

Channel 2202898

Private Public Settings Sharing API Keys Data Import / Export

Modbus RTU

Private Public Settings Sharing API Keys Data Import / Export

ThingSpeak™ Channels Apps Devices Support

New Channel

Name New_channel 2

Description

Field 1 Field Label 1 ☒

Field 2 Field Label 2 ☒ 3

Field 3 ☐

Video URL http://

Show Status ☐

Save Channel 4

Private View Public View Channel Settings Sharing API Keys 5 Data

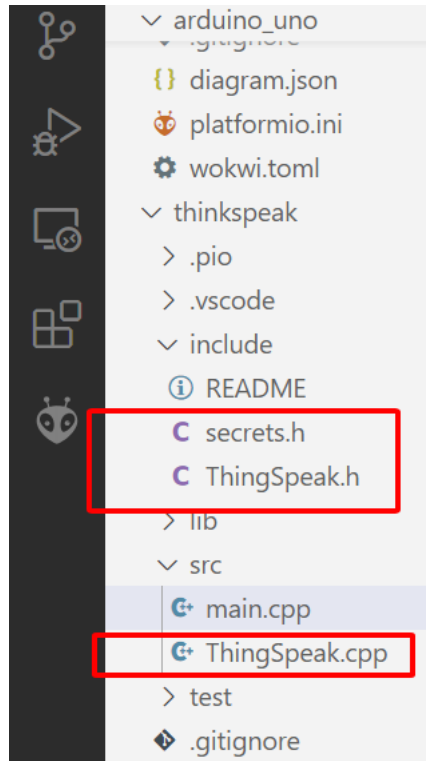
Write API Key

Key 9RYEGXKALIU7W828 5

Generate New Write API Key 6

AP ke A

4.3. Programming cloud connectivity

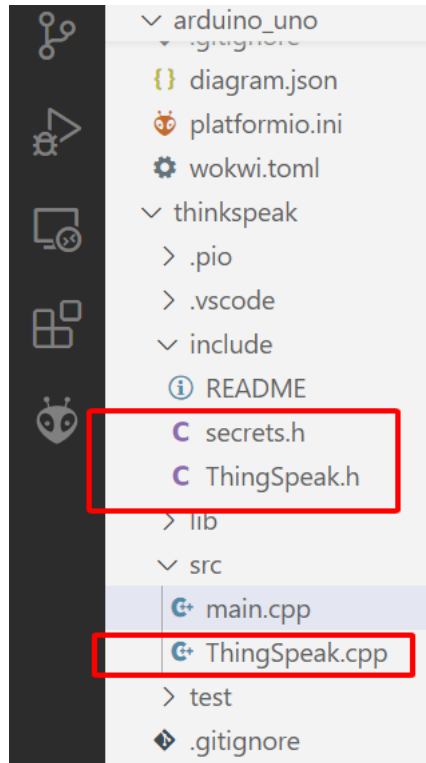


```
1  #include <WiFi.h>
2  #include "ThingSpeak.h"
3  char ssid[] = "Wokwi-GUEST";
4  char pass[] = "";

8  WiFiClient client;
9  unsigned long myChannelNumber = 2923449;
10 const char * myWriteAPIKey = "9RYEGXKALIU7W828";

17 void setup()
18 {
19     Serial.begin(115200);
20     WiFi.mode(WIFI_STA);
21     ThingSpeak.begin(client);
22 }
```

4.3. Programming cloud connectivity

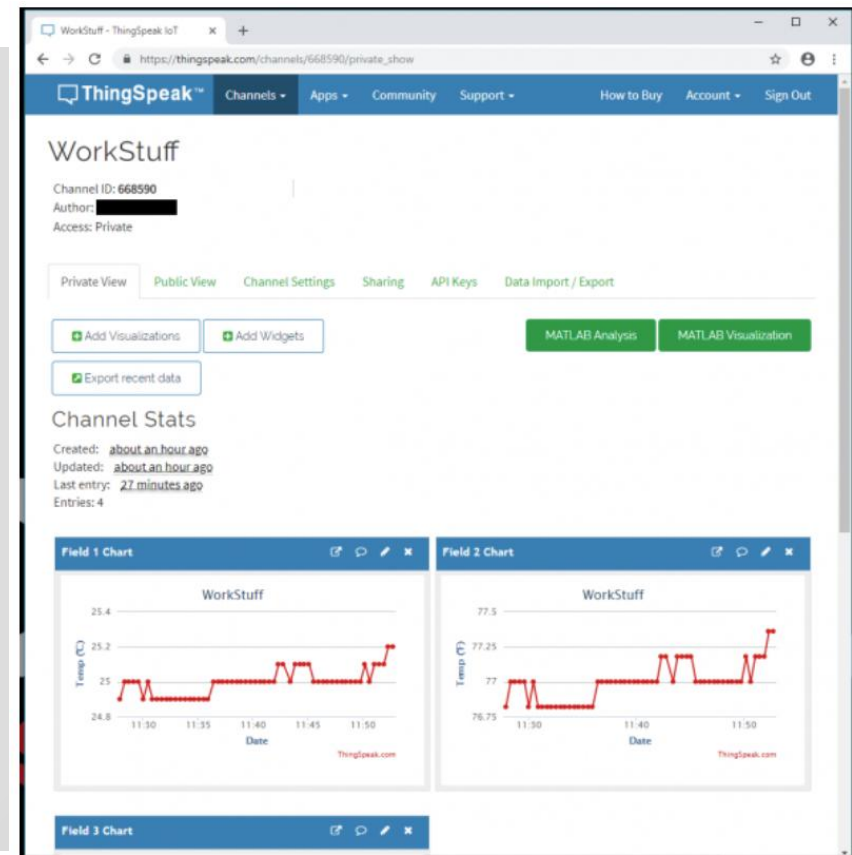
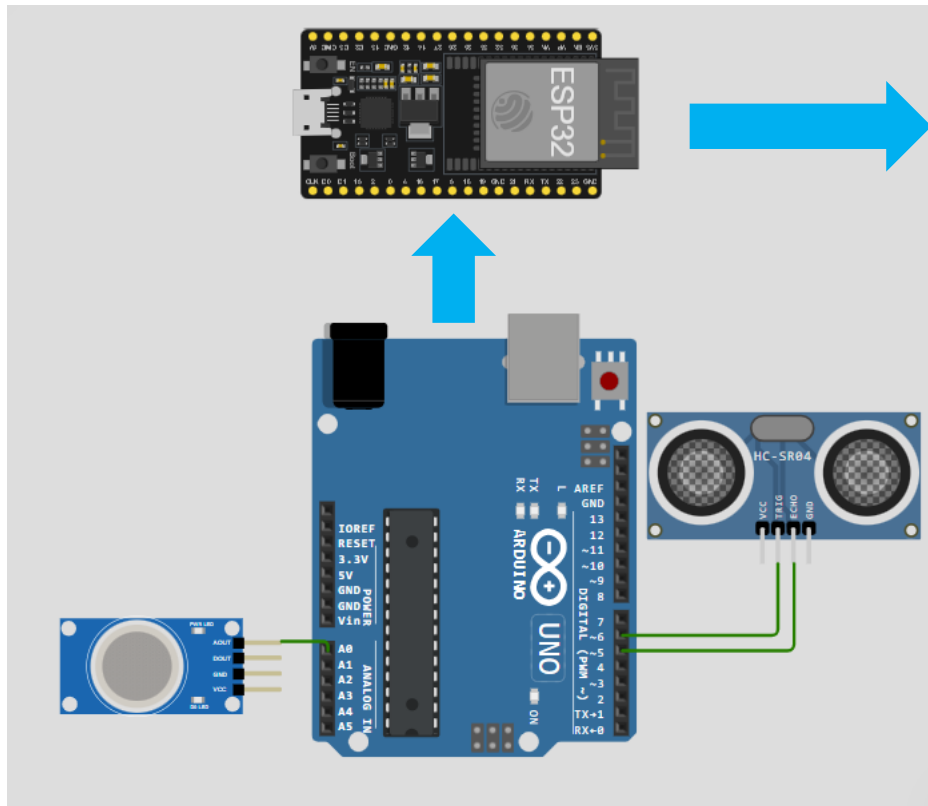


```
25 // Connect or reconnect to WiFi
26 if(WiFi.status() != WL_CONNECTED)
27 {
28     while(WiFi.status() != WL_CONNECTED)
29     {
30         WiFi.begin(ssid, pass);
31         Serial.print(".");
32         delay(5000);
33     }
34     Serial.println("Connected.");
35 }
```

```
39 ThingSpeak.setField(1, number3);
40 ThingSpeak.setField(2, number4);
41 int x = ThingSpeak.writeFields(myChannelNumber, myWriteAPIKey);
42 if(x == 200)
43     Serial.println("Channel update successful.");
44 else
45     Serial.println("Problem updating channel. HTTP error code " + String(x));
```

4.3. Programming cloud connectivity

Project 3:



Thank for attention